

School of Engineering and Technology

Ref. No. CSOET/CSE/INT/2026/114

Dated: 15th July 2026

NOTICE (Test Announcement Batch: 2024)

Course Code: 26CS022		Course Title: Programming Abstractions	
Examination: Sessional Test-I		Max. Marks: 40	Duration: 90 Mins.
Date of Examination: 24 th July 2026	Time: Session-I: 09:30 AM-11:00 AM Session-II: 12:00 PM-01:30 PM	Mode of Examination:	Offline
Programme & Semester: B.E. (CSE)-I			

Syllabus for Test:

Session No.	Topics
1-47	Java Basics: Syntax, JVM Architecture, Variables, Data Types, Type Casting, and Operators, Control Flow: Conditional statements (If-else, Switch), Loops (For, While, Nested), and Jump statements (Break/Continue), Complexity Analysis: Detailed Big-O, Big-Omega, Big-Theta, and Space Complexity analysis for iterative codes, Arrays & I/O: 1D/2D Arrays, Memory allocation, Scanner class, and Fast I/O using BufferedReader, String Handling: String Pool, Immutability, StringBuilder vs StringBuffer, and manipulation methods, Singly Linked List: Node creation, Insertion (Start/End/Middle), Deletion, and Searching logic. Doubly Linked List: Bidirectional Traversal, implementation of DLL, and Pointer management; Circular Linked List: Implementation and use cases (e.g., Round Robin Scheduling), LL Problems: Reversing a List (Iterative & Recursive), LL Problems: Merging two sorted lists. Practice Problems

Pattern of Question Paper:

Mark(s)	1 Mark MCQ	2 Marks MCQ	5 Marks Coding	10 Marks Coding
No. of Questions	10	5	2	1
Options	-	-	-	-

Name of Course Coordinator: Dr. Ashutosh Kumar Dubey

Signature of Course Coordinator